

Architectural Selection & Justification Document

System Name: Pet Adoption & Shelter Management System (PASMS)

Selected Architectural Style: 3-Tier Layered Architecture

1. Architectural Choice

We have selected a 3-Tier Layered Architecture (Presentation Layer, Business Logic Layer, and Data Layer) for the Pet Adoption & Shelter Management System (PASMS). This structural pattern isolates key concerns into distinct horizontal layers, ensuring that user interaction, core business rules, and persistent storage interact through clearly defined interfaces without tight coupling.

2. Scenario-Related Justifications

Clear Separation of Concerns Across Diverse Stakeholders:

PASMS serves two primary stakeholder groups with fundamentally different functional needs—Adoption Applicants searching catalogs and submitting questionnaires, and Shelter Staff updating medical intake records and managing quarantine statuses. Decoupling the Presentation Layer from the underlying logic ensures that frequent updates to user interfaces (such as responsive portal design or catalog display features) can occur independently without risking regression or downtime in critical back-end workflows.

Modularity & Maintainability of Complex Business Logic:

Managing pet adoptions involves multi-step validation workflows, including background screening, medical intake clearance, and foster placement tracking. By isolating these rules inside dedicated business components (Application Manager, Medical Intake Component, and Payment Service), developers can modify specific adoption criteria or medical protocols without refactoring the entire system codebase.

Hardware & External Service Abstraction:

The system needs to interface with external third-party systems like payment processing gateways. A layered structure allows the Payment Service Component to encapsulate API communication details. If the payment provider or protocol

changes in the future, only this isolated business component needs updating, keeping the rest of the application unaffected.

3. Security Advantage

Enforced Layer Isolation and Perimeter Defense:

In a layered architecture, components are restricted to communicating strictly with adjacent layers through exposed provided/required interfaces. The Presentation Layer is completely isolated from the Data Layer. External client requests from web browsers or mobile portals cannot bypass the Business Logic Layer to query or modify the database directly. All incoming requests for sensitive data—such as confidential applicant financial questionnaires or restricted animal medical history—must pass through authentication, authorization, and input-sanitization rules enforced by business components before reaching the Database Component. This setup prevents common web vulnerability vectors such as SQL injection and unauthorized data access.

4. Performance Benefits

Targeted Caching & Optimized Query Execution:

PASMS must satisfy strict non-functional performance requirements, such as handling multi-attribute catalog filtering (by Species, Age, Breed, and Good with Kids) with sub-200ms page response times under peak user load. Isolating data interactions within the Data Layer enables targeted database indexing and query optimization specifically designed for search operations. Furthermore, read-heavy data—such as animal availability listings—can be cached at the Business/Application layer level, significantly reducing database load and response latency.

Scalability & Resource Management:

The layered separation allows for independent optimization and resource allocation. For instance, high read traffic on the public adoption portal during adoption drives can be scaled horizontally at the Presentation Layer without needing to duplicate heavy back-end database resources.